One Line, Two Model Diagnostics

AP Statistics · Topic 2.7 (CED 5.4) · B: 2027-02-11 · E: 2027-04-07 · TEACHER KEY

CED TETHER

- ENDURING UNDERSTANDING: DAT-1 Regression models may allow us to predict responses to changes in an explanatory variable.
- SKILLS:
- **Skill 2.B** — Construct numerical or graphical representations of distributions.
- **Skill 2.A** — Describe data presented numerically or graphically.
- LO DAT-1.E Represent differences between measured and predicted responses using residual plots. [Skill 2.B]
- EK DAT-1.E.1 The residual is the difference between the actual value and the predicted value: residual = y - $\widehat{y}$.
- EK DAT-1.E.2 A residual plot is a plot of residuals versus explanatory variable values or predicted response values.
- LO DAT-1.F Describe the form of association of bivariate data using residual plots. [Skill 2.A]
- EK DAT-1.F.1 Apparent randomness in a residual plot for a linear model is evidence of a linear form to the association between the variables.
- EK DAT-1.F.2 Residual plots can be used to investigate the appropriateness of a selected model.

Essential question: How should correlation and residuals be reconciled when judging a model for safety planning?

DOK-3 skill: 4.B · **FRQ pattern:** `correlation-residuals-model-choice-and-scope`

DOK rationale: Part (c) reconciles correlation with ordered residual locations, connects signed error to a safety decision, and requires both a replacement-model diagnostic and a scope boundary.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK:** 1 → 3 **Minutes:** 45

Teacher does

- Ask which evidence should control the model judgment; do not name the residual form.
- Run the follow-along at minute 5; return at minute 33 for correlation, residual, safety, replacement, and scope evidence. Score part (c).

Students do

- Commit to a judgment, complete the follow-along, finish (a)–(c), revise, and turn in.

Questions to ask

- What does a positive residual say about the prediction?
- Does sign alone reveal where the errors occur?
- What would you require from a replacement model's residuals?

Adult role

Circulate. Require students to reconcile the two diagnostics and connect error direction to the safety use.

[WARN] WATCH FOR

- Using correlation as a substitute for inspecting errors.
- Counting residual signs without checking their locations.
- Reversing underprediction and overprediction.

SCORING PART (c) — E / P / I

Expected elements

- **reconciles-diagnostics** (required) — Uses the U-shaped residual locations despite the large correlation
- **rejects-sign-counting** (required) — Explains why both residual signs do not establish random scatter
- **uses-safety-error** (required) — Links the four-foot underprediction at 50 mph to a safety consequence
- **sets-check-and-scope** (required) — Requires unpatterned replacement residuals and bounds the setting

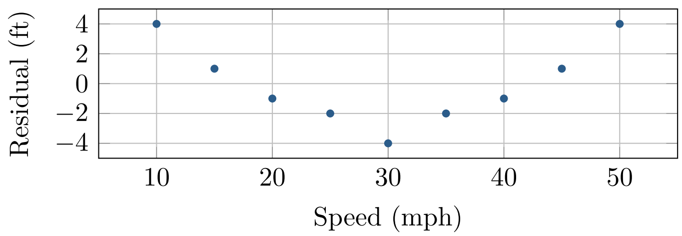
E	Reconciles both diagnostics, rejects sign counting, uses the high-speed error, requires a replacement check, and bounds scope.
P	Identifies the residual pattern but incompletely explains correlation, safety, replacement evidence, or scope.
I	Accepts the line solely from correlation or residual signs, or reverses the meaning of residual sign.

[STAR] TODAY'S PROBLEM — annotated

Suppose a road-safety lab records braking distance y at speeds x from 10 through 50 mph on one vehicle and dry surface. The line is $\hat{y} = -10 + 2.5x$, the correlation is $r = 0.997$, and the residual plot is shown. Actual distances at 10, 30, and 50 mph are 19, 61, and 119 feet.

Ezra: The large correlation and residuals on both sides of 0 justify using the line across the tested speeds.

June: I would use where the residuals occur and the 50-mph error to judge safety, then check any replacement model separately.



FIRST TAKE (5 min)

Which evidence should control the model judgment? Cite one numerical or graphical feature.
Key: Not graded. Students should decide what each diagnostic establishes before judging the line.

Rules callout "READING A RESIDUAL PLOT (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** Read the residuals at 10, 30, and 50 mph and state whether the line underpredicts or overpredicts at each speed.

Key: The residuals are 4, -4, and 4 feet. Positive residuals at 10 and 50 mph mean underprediction; the negative residual at 30 mph means overprediction.

DOK 2 **(b)** Verify those residuals from the equation and actual distances. Describe the full residual plot, including its form and where the line tends to underpredict or overpredict.

Key: Predictions at 10, 30, and 50 mph are 15, 65, and 115 feet, so $19 - 15 = 4$, $61 - 65 = -4$, and $119 - 115 = 4$. The residuals form a U-shaped sequence: positive at 10–15, negative at 20–40, then positive at 45–50 mph. The line underpredicts at the ends and overpredicts in the middle.

DOK 3 **(c)** Evaluate both judgments by reconciling $r = 0.997$ with the residual locations. Assess what the large correlation and both residual signs do or do not establish, use the 50-mph error for a safety consequence, specify evidence required before trusting a replacement model, and bound the conclusion to the observed setting.

Key: June's approach is warranted. The value $r = 0.997$ describes very strong positive linear association in the original variables, but the U-shaped residual sequence shows systematic error left by the line. Both signs do not imply random scatter when their locations are predictable. At 50 mph the line predicts 115 feet but the observed distance is 119 feet, a 4-foot underprediction that could make a stopping zone too short. Fit a plausible curved model, require unpatterned residuals around 0, and validate it on new trials. Conclusions remain limited to this vehicle, dry surface, and 10–50 mph.

EXIT

One thing the video changed about my first take: ____________________